

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1

Hour

QUESTIONS: 4

Total: Marks:40

Annexure A

5

Code of Conduct:

I __S.Jotheeswaran_____ (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Jotheeswaran

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

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Industry Problem Task: Healthcare AI Solution

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach and Features

Inductive learning is a machine learning paradigm where an algorithm attempts to formulate a generalized rule or function by observing specific instances. In this medical context, the inductive learning approach is Supervised Binary Classification. The model is presented with a training set of labeled patient records (X) to approximate a target function that maps these physiological features to a specific diagnostic outcome (Y).

Target Variable: Diabetes Status (Categorical/Binary: 1 = Positive/Diabetic, 0 = Negative/Non-diabetic). This is the dependent variable we are attempting to predict.

Top 3 Key Features & Clinical Justification:

- 1. Glucose Level: This is statistically the most deterministic feature. Diabetes is fundamentally a disease characterized by impaired insulin production or utilization, resulting in elevated blood glucose. A fasting glucose level above 126 mg/dL is a primary clinical diagnostic criterion.
- 2. BMI (Body Mass Index): High BMI (obesity) is a well-documented primary risk factor for Type 2 Diabetes. Excess adipose tissue causes cells to become resistant to insulin, directly driving the pathophysiology of the disease.
- 3. Family History: Diabetes has a strong genetic component. This feature acts as a proxy for genetic predisposition and inherited metabolic risks that are not captured by immediate, observable vital signs like blood pressure.

(b) Handling Class Imbalance

Medical datasets are heavily skewed, naturally containing more healthy patients than diseased ones. Training on imbalanced data causes algorithms to become biased towards the majority class, optimizing for overall accuracy by simply ignoring the minority class (the 'Accuracy Paradox').

Proposed Strategy: SMOTE (Synthetic Minority Over-sampling Technique) combined with Stratified Cross-Validation.

Justification: Unlike random under-sampling, which throws away valuable data (reducing the model's ability to learn variance in the healthy population), and unlike random over-sampling, which duplicates exact records (leading to severe overfitting), SMOTE is structurally superior. It works by selecting minority examples that are close in the feature space, drawing a line between them, and generating new, synthetic data points along that line. This expands the decision boundary of the diabetic class organically. Alternatively, applying Algorithmic Class Weights (e.g., assigning a 10x higher penalty for misclassifying a diabetic patient) is an excellent non-destructive strategy.

(c) Dataset Splitting and Preprocessing

Split Ratio: 70:30 (Training:Testing) using Stratified Splitting.

Justification: A 70% split allocates sufficient data for the algorithm to learn the underlying variance and complex interactions between variables. The 30% testing set is crucial for evaluating generalization. Stratification ensures that the original ratio of diabetic to non-diabetic patients is perfectly maintained in both sets, preventing a scenario where the test set accidentally contains zero diabetic patients.

Preprocessing Steps Applied:

- Normalization (StandardScaler): Applied to continuous variables like Glucose, BMI, and Blood Pressure. It shifts the data to have a mean of 0 and a standard deviation of 1 (Z-score normalization). Impact: Distance-based algorithms or gradient descent optimizers will converge much faster, and features with naturally large ranges (like Glucose, 0-200) will not mathematically overpower features with small ranges (like BMI, 15-40).
- One-Hot Encoding: Applied to categorical data. Impact: It transforms categorical strings into binary matrices (0s and 1s), ensuring the algorithm does not interpret arbitrary numeric labels as having an ordinal relationship (e.g., assuming Category 3 is 'greater' than Category 1).

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Structure and Root Node

A standard induction using the CART (Classification and Regression Trees) algorithm would likely result in the following top hierarchy:

- Level 0 (Root Node): Glucose Level ≤ 125.5 mg/dL
- Level 1 (Left Child - Low Glucose): BMI ≤ 29.5 (Class: Non-Diabetic)
- Level 1 (Right Child - High Glucose): Family History == True (Class: Diabetic)
- Level 2: Splits based on Blood pressure or Age to resolve remaining mixed nodes.

Root Node Justification: The Decision Tree algorithm iteratively evaluates every feature to find the split that maximizes Information Gain (or minimizes Gini Impurity). Glucose is selected as the root node because splitting the patients based on glucose levels creates two child nodes that are the most 'pure' (homogeneous). It separates the vast majority of

healthy patients from sick patients immediately, offering the highest mathematical reduction in uncertainty.

(b) Evaluation and False Negatives

Metrics: Accuracy measures overall correctness. Precision measures how many predicted diabetics actually have it. Recall (Sensitivity) measures how many actual diabetics were successfully identified. The F1-Score provides a harmonic mean of Precision and Recall.

The False Negative Problem: A False Negative is a 'Miss'—the model predicts a patient is healthy when they actually have diabetes. In a clinical diagnostic setting, a False Negative is catastrophic. It leads to missed treatments, disease progression, and severe complications (neuropathy, blindness).

Reduction Strategy: To minimize False Negatives, we must maximize Recall. Clinically, we justify trading Precision for Recall. We can achieve this by adjusting the decision threshold. Instead of requiring a >50% probability to classify as Diabetic, we lower the threshold to >20%. This makes the model 'hyper-sensitive', flagging any patient with mild risk factors. This will increase False Positives (healthy patients getting flagged), but clinically, the cost of a follow-up blood test (False Positive) is infinitesimally smaller than the cost of a missed diabetes diagnosis (False Negative).

(c) Post-Pruning (Cost-Complexity Pruning)

Decision Trees are highly prone to overfitting; if left unconstrained (pre-pruning phase), they will grow until every leaf node contains 1 sample, essentially memorizing the training data and noise.

Post-pruning works by growing a massive tree and then systematically collapsing branches that contribute the least to overall accuracy, controlled by a complexity parameter (α). It penalizes the tree for having too many nodes.

Comparison & Deployment: The unpruned tree will have ~100% training accuracy but poor testing accuracy. The post-pruned tree will have slightly lower training accuracy (~85%) but much higher testing accuracy (~82%), demonstrating robust generalization. For clinical deployment, the post-pruned model is vastly superior. Not only does it generalize better to new patients, but a shallower tree is highly interpretable. Doctors can trace the logic of a 4-level tree and verify its medical validity, whereas a 20-level tree acts as an opaque, uninterpretable black box.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression Comparison

Logistic Regression is a parametric statistical model that uses a sigmoid function to map linear combinations of features into a probability score between 0 and 1.

Performance Comparison: A Decision Tree creates complex, non-linear, orthogonal step-boundaries. Logistic regression creates a single, smooth, linear decision boundary. On complex, highly interactive data, a tuned Decision Tree might yield a higher raw Accuracy. However, Logistic Regression is generally superior for the AUC-ROC (Area Under the Receiver Operating Characteristic Curve) metric because its probability outputs are inherently well-calibrated, whereas unpruned Decision Trees output brittle probabilities (often 0% or 100%).

Clinical Preference: Logistic Regression is heavily preferred for Risk Stratification. Instead of a binary 'Yes/No' (from a tree), Logistic Regression tells a doctor: 'This patient has a 72% probability of developing diabetes.' Furthermore, it provides interpretable coefficients (e.g., 'Every 1-unit increase in BMI increases the log-odds of diabetes by 0.4'), allowing clinicians to quantify exact risk factors.

(b) Feature Importance Agreement

Top 3 Predictors: Glucose Level, BMI, Age (or Family History).

Do they agree?: Generally, yes, both models will identify the same top predictors, but they calculate them differently.

- Decision Tree: Importance is calculated by tracking how much Gini Impurity is reduced every time a feature is used to split a node. Glucose will be highest because it's at the root.

- Logistic Regression: Importance is evaluated by looking at the absolute magnitude of the learned coefficients (weights) after the data is normalized. Glucose will have the highest positive weight.

Justification: The underlying biological reality dictates the correlations. Regardless of the algorithm's mathematical mechanics (information theory vs. maximum likelihood estimation), the strongest physiological signals (Glucose and BMI) will dominate the predictive landscape.

Task 4: Reinforcement Learning for Treatment

(a) Markov Decision Process (MDP) Modeling

We frame chronic disease management as a sequential decision-making problem over time (an MDP):

- - States (S): A representation of the patient's current health status. e.g., $S = \{\text{Healthy, Pre-Diabetic, Type-2 Managed, Type-2 Unmanaged}\}$.
- - Actions (A): Clinical interventions available to the agent. $A = \{\text{Recommend Diet, Prescribe Exercise, Prescribe Metformin, Order Blood Test Monitor}\}$.
- - Reward (R): The scalar feedback signal. e.g., +50 for returning to 'Healthy', -100 for transitioning to 'Unmanaged' (hospitalization), -1 for the cost/side-effects of medication, -0.1 for the effort of diet/exercise. This incentivizes the agent to find the least invasive path to health.

- - Transition Dynamics (P): The transition probability $P(S' | S, A)$. E.g., there is a 60% probability that a 'Pre-Diabetic' patient transitions to 'Healthy' if given the 'Exercise' action, but only a 10% chance if given the 'Monitor' action. This represents biological uncertainty.

(b) Q-Learning Application and Convergence

Q-Learning iteratively updates the expected long-term utility of taking an action in a state using the Bellman Equation: $Q(s,a) = Q(s,a) + \alpha * [Reward + \gamma * \max(Q(s',a')) - Q(s,a)]$.

Example Q-Table Update (Assume $\alpha=0.1$, $\gamma=0.9$, Init $Q=0$):

Iteration 1 (Episode 1):

1. State: Pre-Diabetic. Action: Exercise. Transition: Healthy. Reward: +50.
Update: $Q(\text{Pre-Diabetic}, \text{Exercise}) = 0 + 0.1 * [50 + 0.9(0) - 0] = +5.0$
2. State: Type-2 Unmanaged. Action: Diet. Transition: Type-2 Unmanaged. Reward: -100.
Update: $Q(\text{Unmanaged}, \text{Diet}) = 0 + 0.1 * [-100 + 0.9(0) - 0] = -10.0$

Iteration 2 (Episode 2):

3. State: Pre-Diabetic. Action: Exercise. Transition: Healthy. Reward: +50.
Update: $Q(\text{Pre-Diabetic}, \text{Exercise}) = 5.0 + 0.1 * [50 + 0.9(\max Q \text{ in Healthy}) - 5.0]$. The value continues to grow, reinforcing this action.

Convergence Criterion: The Q-Learning algorithm is considered converged when the maximum change in any Q-value across the entire matrix (L-infinity norm) between episodes falls below a vanishingly small threshold (epsilon, e.g., 0.001). This mathematically indicates that the agent has stabilized its policy and further exploration yields no new knowledge.

(c) Final Policy and Medical AI Ethics

Extraction: The learned policy (π^*) is deterministic. For any given patient state, the agent simply looks up the row in the Q-table and selects the column (Action) with the highest numeric value.

Ethical Considerations in Clinical RL Deployment:

- - Patient Safety (Safe Exploration): Standard RL requires the agent to explore 'bad' actions to learn they are bad. We cannot allow an AI to trial random lethal drug dosages on human patients just to update a Q-table. The agent must be trained in a simulated environment first, and bounded by strict safety constraints (Safe RL) that forbid medically contraindicated actions.
- - Algorithmic Bias: If the MDP transition probabilities are derived from hospital data serving predominantly wealthy demographics, the agent's recommended policy might

fail when deployed in lower-income areas where adherence to expensive diets or medications differs due to socioeconomic factors.

- - Explainability & Trust: RL models are inherent black boxes. An agent might recommend halting medication because it calculated a complex, long-term statistical benefit. A physician cannot ethically execute this recommendation without understanding **why**. The lack of causal explainability is a major regulatory barrier for AI in healthcare.